GG-M-00114b(GSA-FSS)
July 24, 1963
Revision of
Int. Fed. Spec. GG-M-00114
March 7, 1963

INTERIM FEDERAL SPECIFICATION

MARKER, TUBE TYPE, FELT TIP, (CAPILLARY-ACTION)

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Standardization Division, Washington 25, D. C., based upon currently available technical information (see Sec. 6, paragraph 6.2). It is recommended that Federal agencies use it in procurement and forward any recommendations for changes to the preparing activity at address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers tube-type felt-tip (capillary action) markers for use in freehand marking, writing, drawing and identification marking of various types of material.

1.2 Classification.

1.2.1 Type. Marker shall be of but one type.

1.2.2 Colors of ink. (See 3.5.)

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on the date of invitation for bids, form a part of this specification:

Federal Specifications:

UU-P-121 - Paper, Bond and Writing, White and Colored.
UU-T-110 - Tape, Paper, Computing-Machine.
CCC-T-191- Textile Test Methods.
PPP-B-566- Boxes; folding, Paperboard.
PPP-B-636- Box, Fiberboard.
PPP-B-665- Boxes; Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676- Boxes; Set-Up, Paperboard.
PPP-F-320 - Fiberboard; Sheet, Stock and Cut Shapes.
QQ-A-561 - Aluminum Alloy Plate and Sheet, 1100

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

Federal Standards:

Fed. Std. No. 102 - Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

FOR INSTRUCTIONS ON OBTAINING COPIES OF FEDERAL SPECIFICATIONS AND STANDARDS, SEE LAST PAGE.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

Governmental:

Federal Hazardous Substances Labeling Act.

(Copies of the above publication may be obtained upon application, accompanied by money order, coupon, or cash, to the Superintendent of Documents, U.S. Government Printing Office, Washington 25, D.C.).

3. REQUIREMENTS

3.1 Bid samples Each bidder shall submit with his bid 12 markers of the color he proposes to furnish, for the purpose of testing for compliance with the requirements of this specification prior to award.

3.2 Material. The manufacturer is given wide latitude in his selection of materials and processes of manufacture provided the product meets the requirements of this specification.

3.3 Design. The marker shall be designed to operate satisfactorily by capillary action. The ink shall be held within the marker by an absorbent material so that there is no free liquid within the marker.

3.4 Composition of ink. The marker ink shall consist essentially of a dye dissolved in an organic solvent. Binding ingredients to provide surface adhesion shall be added.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

3.5 Colors of ink. Markers shall be furnished with ink of the following colors, as specified in the invitation for bids, contract, or order (see 6.1): Black, red, blue, green, orange, yellow, brown and purple.

3.5.1 Markings. Markings made as described in 4.3.1.1 shall be of the color specified and clearly legible.

3.6 Hazardous substances. The marker ink shall contain no benzol (benzene), aniline oil, fusel oil, chloronated compounds, or hydrolyzable chlorine derivatives. In addition, themarker shall comply with the regulations of the Federal Hazardous Substances Labeling Act. A certificate of compliance with both of the requirements of this paragraph shall be furnished by the bidder.

3.7 Writing nib. The writing nib shall conform to the shape and dimensions shown in figure 1, and shall be permanently secured in position without excessive compression which might affect the capillary flow of ink.

3.8 Drying time, resistance to water, light fastness, and non-transferability.

3.8.1 Drying time. Markings on panels prepared as described in 4.3.1.1 shall not smear, and the writing paper which is drawn across the markings shall not be smudged, when tested as specified in 4.3.1.2. A slight trace of color transferred to the writing paper is permissible when testing all panels. Slight striation of the marking on aluminum panel is permissible.

3.8.2 Resistance to water. Markings on panel prepared as described in 4.3.1.1 shall not show a marked change in color or shade, and the writing paper drawn over the markings shall not be smudged when tested as specified in 4.3.1.3. A slight trace of color transferred to the writing paper is permissible.

3.8.3 Light fastness. Markiogs on the exposed specimen prepared as described in 4.3.1.4 shall not show a marked change in color or shade when tested as specified in 4.3.1.4.

3.8.4 Non-transferability. Markings on panel prepared as described in 4.3.1.1 shall not transfer to the cover paper when tested as specified in 4.3.1.5. A slight trace of color is permissible.

3.9 Accelerated aging. Marker shall not leak, and markings shall not show a marked change in color or shade when tested as specified in 4.3.2. A slight variation in mark intensity due to felt pattern is permissible.

3.10 Recuperation. The marker shall produce a continuous mark not less than 0.094 inch wide within 4 strokes after standing with cap off for 30 minutes, within 6 strokes after standing with cap off for 60 minutes, and within 10 strokes after standing with cap off 8 hours and recuperating for 24 hours with cap on as specified in 4.3.3. Slight striation or mottling of the mark is permissible.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

3.11 Write-out distance. The marker shall write a continuous line of the color indicated, 0.125 + 0.031 inch wide, for the following minimum distance, when tested as specified in 4.3.4:

Color	Minimum linear footage
Black	2475
Blue	2025
Red	2025
Green	2025
Orange	2025
Yellow	2025
Brown	2025
Purple	2025

3.11.1 Standard performance samples. When tested as specified in 4.3.4, the line at the beginning of the write-out shall be at least equal to standard performance sample No. 1. The line at the end of the required minimum distance shall be at least equal to standard performance sample No. 2. (See 6.3)

3.12 Body and cap tightness. The marker body shall be so constructed and the cap so seated as to prevent loss of ink by evaporation or leakage. The cap shall be easily removed without the use of tools. The average weight loss per marker shall not exceed 3.0 milligram when tested as specified in 4.3.5.

3.13 Finish. Markers shall have a smooth, even surface. The external surface shall be finished in accordance with standard commerical practice. The cap or body of the marker shall be colored in a manner to identify color of ink content.

3.14 Identification marking. Markers shall be marked in a legible manner with the manufacturer's name or trademark of such known character that the source of manufacture can be readily determined.

3.15 Shelf life. The markers, under normal storage conditions, shall retain their serviceability as defined by this specification for not less than 12 months from time of delivery. Suppliers shall make replacement, without cost to the Government, of all markers failing to meet this requirement.

3.16 Workmanship. Markers shall be free from imperfections or defects, which might affect their serviceability or durability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES.

4.1 Supplier's responsibility. Unless otherwise specified (see 6.1), the supplier is responsible for the performance of all inspection requirements as

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling and examination.

4.2.1 Size of lot. A lot shall consist of all markers of the same color of ink offered for delivery at one time.

4.2.2 Sampling for and examination of packaging, packing and marking for shipment. Sample shipping containers of packaged and packed markers shall be taken from an assembled lot for examination for compliance with the requirements of Section 5 of this specification with respect to correctness of material and construction of packaging and packaging and packing; adequacy of closure; and correctness, completeness, and legibility of marking and labeling. The sample size shall be in accordance with Military Standard MIL-STD-105. The sample unit shall be one shipping container. The acceptable quality level (AQL) shall be 4.0 percent defective.

4.2.3 Sampling for visual examination of markers. A randon sample of markers shall be taken from packages contained in each shipping container obtained in accordance with 4.2.2. The sample unit shall be one marker. The sample size shall be in accordance with Inspection Level L-4 of Military Standard MIL-STD-105.

4.2.4 Visual examination of markers. Visual examinations of markers shall be in accordance with the classification of defects listed in table 1. The acceptable quality level (AQL) shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

Table I. Classification of defects

Defects	Category	
	Major	Minor
Ink color not as specified	x	
Nib size not as specified	x	
Nib shape not as specified	x	
Evidence of leakage	x	
Cap Loose	x	
Finish not smooth or even		x
Identification marking not as specified		x
Workmanship: Any defects which --		
Impair serviceability	x	
Affect appearance		x

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

4.2.5 Sampling for tests of markers.

4.2.5.1 Drying time, resistance to water, light fastness, non-transferability, accelerated aging, recuperation and write-out distance. Sampling for the drying time (4.3.1.2), resistance to water (4.3.1.3) light fastness (4.3.1.4), non-transferability (4.3.1.5), accelerated aging (4.3.2), recuperation (4.3.3) and write-out distance (4.3.4) tests shall be in accordance with level L-2 of Military Standard MIL-STD-105. The sample unit shall consist of all the markers needed to conduct all the tests one time. For the foregoing tests, the sample unit shall be 4 markers.

4.2.5.2 Body and cap tightness. A random sample of 5 markers shall be selected for the body and cap tightness test (4.3.5).

4.2.6 Examination of test results.

4.2.6.1 Drying time, resistance to water, light fastness, non-transferability, accelerated aging, recuperation and write-out distance. Sample markers selected in accordance with 4.2.5.1 shall be subjected to the tests specified in 4.3.1.2, 4.3.1.3, 4.1.1.4, 4.3.1.5, 4.3.2, 4.3.3 and 4.3.4. Failure to pass one or more of the tests shall render the sample unit defective. The acceptable quality level (AQL) shall be 4.0 percent defective.

4.2.6.2 Body and cap tightness. Sample markers selected in accordance with 4.2.5.2 shall be subjected to the test specified in 4.3.5. Failure to pass this test shall be cause for rejection of the lot.

4.3 Tests. Tests, unless otherwise specified hereinafter, shall be conducted in an atmosphere of 70° ± 10° F. and at a relative humidity of 50 ± 10 percent.

4.3.1 Drying time, resistance to water, light fastness, and non-transferability.

4.3.1.1 Preparation of test panels. Markings in the color specified, each 6 inches in length, shall be made on paper, fiberboard, and aluminum panels. Paper, when specified hereinafter for tests, except in 4.3.4 (see 4.3.4.1), shall be white writing paper, 32 pound (17 x 22-1000 sheets), Type V, conforming to Specification UU-P-121. Fiberboard, when specified hereinafter for tests, shall be type SF, class weather resistant, grade V3s, conforming to Specification PPP-F-320. Aluminum, when specified hereinafter for tests, shall be 1100 aluminum alloy flat sheet, temper H14, thickness approximately 1/16 inch, conforming to Specification QQ-A-561.

4.3.1.1.1 Aluminum panels. (Cleaning). The aluminum panels, approximately 4x6 inches, shall be abraded with a fine aluminum oxide cloth (emery) No. 320 until a polished surface free from corrosion defects is obtained. The abrading strokes shall be made in a direction parallel to the length of the panel. The panel shall then be degreased by immersing it in toluene at 70° ± 1° C for 10 minutes, removed and allowed to dry. The panel shall then be immersed in absolute alcohol at 50° ± 1° C for 10 minutes, removed and allowed to dry. The panel surface shall not be touched prior to testing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

4.3.1.2 Drying time. After marking panels as specified in 4.3.1.1, allow paper and fiberboard panels to dry for 5 seconds; aluminum, 1 minute. Immediately after the specified drying time, place a sheet of paper (see 4.3.1.1) over the markings on the respective panels; place a 200 gram weight, approximately 1-1/4 inches in diameter, uniformly on one edge of the paper; and draw the sheet of paper and the weight perpendicularly across the center of the markings at a rate of approximately 4 inches per second. Examine the markings and paper for compliance with 3.8.1.

4.3.1.3 Resistance to water. Prepare a paper panel (see 4.3.1.1) and allow markings to dry for 5 minutes. Immerse panel in water to a depth correspnding to two-thirds of the height of the markings, for 4 hours. Remove panel at end of immersion period, allow to dry for 1 hour, and then test for smudging as described in 4.3.1.2. Examine the markings and paper for compliance with 3.8.2.

4.3.1.4 Light fastness. Prepare fiberboard panel (see 4.3.1.1) and then allow markings to dry for 1 hour. Two 2-1/2 by 3 inch test specimens, each containing one-half of the marking specified in 4.3.1.1, shall be cut from the panel. One specimen shall be exposed to light as described in method No. 5660 of Specification CCC-T-191 and the other retained but not exposed. Exposure periods shall be as follows: Black markings, 24 hours, red, orange, yellow, brown, and purple, 12 hours; and blue and green, 8 hours each. The two specimens shall then be compared and the exposed specimen examined for compliance with 3.8.3.

4.3.1.5 Non-Transferability. Prepare a paper panel (see 4.3.1.1) and allow markings to dry for 5 minutes. Immediately after the specified drying time the markings shall be covered with three layers of paper (see 4.3.1.1) and a rectangular 4 by 6 inch piece of 1/4 inch plate glass shall be placed flat on top of the test assembly. The entire assembly shall then be placed on a flat surface in a convection-type oven and maintained at a temperature of 80° ± 1° C. for 1 hour. The assembly shall then be removed from the oven and the cover paper examined for compliance with 3.8.4.

4.3.2 Accelerated aging. An unused marker shall be uncapped and the writing nib drawn across a sheet of paper (see 4.3.1.1) for a distance of 8 inches. The marker shall then be recapped tightly, placed in a convection-type oven and maintained at a temperature of 60° ± 1°C for 7 days. At the end of the aging period the marker shall be removed from the oven and immediately examined for leaks. The marker shall then be placed on its side and allowed to cool at room temperature for 24 hours. The cap shall then be removed and the writing nib again drawn across the paper parallel to the existing mark. Examine the markings for compliance with 3.9

4.3.3 Recuperation. One unused sample marker shall be placed with the writing nib up and with cap removed. At 30- and 60-minute intervals the marker nib, positioned as specified in 4.3.4, shall be drawn across a sheet of paper

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

(see 4.3.1.1_ at a rate of approximately 4 inches per second and with a writing pressure of approximately 200 grams, in increments of 8 inches until a continuous mark 0/094 inch wide is attained. Examine markings for compliance with 3.10. Upon completion of the 60-minute markings, the marker shall be placed with the writing nib up and with the cap removed for an additional 7 hours at which time the cap shall be replaced. The marker shall then be placed on its side for 24 hours. The writing nib shall then be drawn across a sheet of white writing paper as specified above and the markings examined for compliance with 3.10.

4.3.3 Write-out distance. An unused sample marker shall be tested on a write-out apparatus suitable for conducting the following test:

The marker body shall be suspended, nib down, in a fixed position perpendicular to the specified paper tape (see 4.3.4.1); shall be turned so that the longest exposed side of the nib (1/4" ± 1/32") faces in the direction of the paper movement after the paper has passed under the marker; and shall have a spring applied pressure of 100 grams for the initial writing pressure. The apparatus shall draw the paper across a flat metal surface under the marker at a rate of 40 + 2 feet per minute. At approximately 225 linear feet the apparatus shall be stopped, the marker capped and allowed to stand nib down for a period of 15 minutes, when the initial writing pressure shall be increased by 10 grams. This procedure shall be repeated in 225-foot segments until the marker fails, at any point within a segment to produce a line 0.094 inch wide or a line which is not at least equal to Standard Performance Sample No. 2. The apparatus shall then be stopped and the linear footage checked for compliance with 3.11.

4.3.4.1 Paper tape used in the above test shall conform to type 1 of Specification UU-T-110.

4.3.5 Body and cap tightness. Five unused sample markers shall be tested as follows:

Uncap and recap each marker to a normal hand tightness 20 consecutive times. Uncap markers and draw a 3-foot linear mark with each sample, and then recap the markers tightly. Immediately take and record the total weight of the five markers. Then allow the markers to stand, tightly capped, for a period of 24 hours. At the end of this period, again take and record the total weight of the five markers. Divide the difference between the two total weights recorded by five. The resulting figure shall be considered the average weight loss per marker. Check this weight loss for compliance with 3.12.

5. PREPARATION FOR DELIVERY

The definitions and applications of levels of packaging and packing for civil agencies shall be as defined in Federal Standard No. 102.

5.1 Packaging. (See 6.1)

5.1.1 Level A. Level A packaging, unless otherwise specified (see 6.1), shall be as required in 5.1.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

5.1.2 Level B. Twelve (12) markers of one color shall be packaged in a snug-fitting folding or set-up paperboard box conforming to Specification PPP-B-566, PPP-B-665, or PPP-B-676. Glass-bodied markers shall be separated by full-depth slotted partitions to provide additional protection.

5.1.3 Level C. Unless otherwise specified, markers shall be packaged in accordance with standard commercial practice.

5.2 Packing. (see 6.1)

5.2.1 Level A. Level A packing, unless otherwise specified (see 6.1) shall be as required in 5.2.2, except that boxes shall be class 2.

5.2.2 Level B. Thirty-six paperboard boxes as specified in 5.1.2, (432 markers), shall be packed in a snug-fitting fiberboard box conforming to Specification PPP-B-636, type I or II, class 1, style RSC. Burst test of fiberboard shall be not less than 275 pounds p.s.i. Closure shall be in accordance with the appendix to the box specification.

5.2.3 Level C. Unless otherwise specified, markers shall be packed in shipping containers of the type, size, and kind commonly used for the purpose in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the carrier rules and regulations applicable to the mode of transportation.

5.3 Marking. (See 6.1)

5.3.1 Civil agencies. In addition to any special marking specified in the contract or order, marking shall be in accordance with Fed. Std. No. 123. Each box of twelve (12) markers shall be marked on one end with the Federal Stock Number, Federal item identification, including color and the quantity contained therein.

5.3.2 Military agencies. In addition to any special marking specified in the contract or order, shipments shall be marked in accordance with Standard MIL-STD-129/ MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein (see 5.1,1, 5.1.3, 5.2.1, and 5.2.3) and procurement documents should specify the following (see 3.1, 3.6 and 3.15):

(a) Title, number, and date of this specification.
(b) Color of markers required (see 3.5)
(c) Inspection responsibility (for civil agencies only) if other than as specified herein; and administrative provisions for inspection records, when applicable (4.1)
(d) Levels of packaging and packing and any special marking for shipment required (see 5.1, 5.2, and 5.3)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

6.2 This Interim Federal Specification represents action in the development of a Federal Specification for tube-type felt-tip (capillary action) markers (see 1.1), which will reflect current manufacturing practices consistent with the needs of Federal agencies. The major change effected herein consists of the addition of the following colors: Orange, yellow, brown and purple.

6.3 Standard Performance samples. Standard performance samples of the minimum acceptable mark at the beginning and end of the write-out are available for inspection at the Quality Control Division Offices in all General Services Administration Regional Offices.

6.4 Federal specifications do not include all types, classes, grades, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

NOTICE: See last page.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

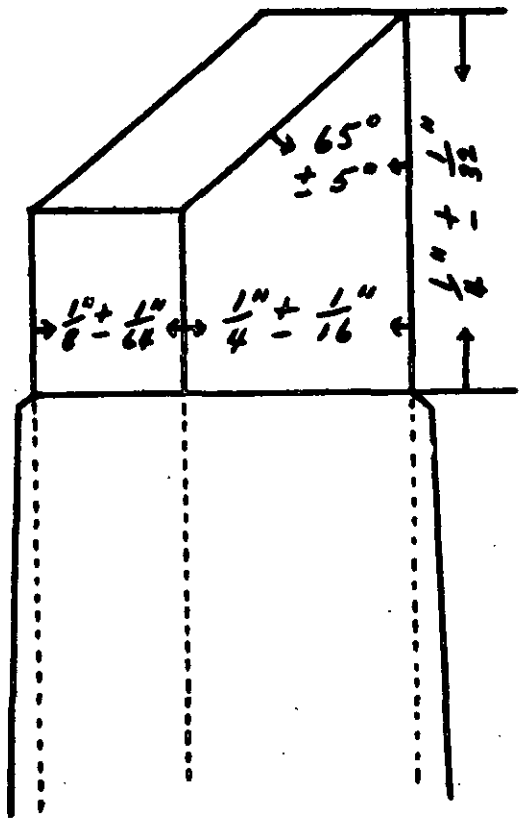


Figure 1. - Writing nib

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

FORM 1 APR 62 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15418

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.